GRC REPORT

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Task Title: Security Operation Center (SOC)
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Introduction

As part of my cybersecurity internship, I participated in a SOC (Security Operations Center) sprint designed to simulate real-world defensive security operations. The sprint aimed to provide hands-on experience with core security practices such as log analysis, incident investigation, malware triage, and vulnerability management.

Our team was tasked with proactively monitoring the organization's infrastructure for threats and responding to incidents following standard frameworks. We were structured into functional sub-teams and collaborated on high-impact activities, including malware analysis, suspicious sign-in investigations, and incident playbook creation. The simulation helped us build our understanding of how a real-world SOC functions, the importance of accurate detection and triage, and how frameworks like NIST support incident handling.

Summary of Key Tasks

During the sprint, our team focused on several critical cybersecurity domains and practical exercises, some of which are detailed below:

a. Suspicious Sign-In Investigation

We were provided with authentication and activity logs relating to a staff member named "Erwin Smith." The investigation focused on identifying login anomalies based on IP geolocation, frequency, and behavior. After log parsing and IP reputation checks, we discovered that while one IP belonged to the UK (and matched expected behavior), three others were from foreign locations (Seychelles, Beijing, Bangkok) and showed signs of malicious intent. This helped simulate real-world threat hunting and emphasized how vital it is to understand normal vs. abnormal account behavior.

b. Malware Analysis ()sing tools like VirusTotal)

Our team examined suspicious IPs and artifacts via VirusTotal to analyze potential malware signatures or malicious indicators. We practiced uploading and scanning files, URLs, and IP addresses to detect whether they were known threats. This enhanced our understanding of threat intelligence enrichment and IOC (Indicator of Compromise) gathering.

c. Vulnerability Management

We reviewed system configurations and vulnerability reports for TesTech's internal services. The tasks involved identifying CVEs affecting critical systems, understanding CVSS scoring,

and suggesting remediation priorities. The vulnerability management workflow emphasized that patch management and proactive scanning are key to reducing the attack surface.

d. Incident Playbook Creation (Suspicious Sign-Ins)

I was selected, alongside another team member, to create a dedicated incident response playbook for suspicious sign-ins affecting TesTech's cloud and internal systems. We used the NIST Incident Response and the playbook included threat detection methods, user/IP profiling, logging practices, training, and alert triage workflows. It was tailored to TesTech's profile as a manufacturing organization with global connectivity.

Lessons Learned

This sprint gave me practical insight into the technical and organizational processes behind incident response. The key lessons I took away include:

Value of Threat Intelligence

Using tools like VirusTotal and WHOIS, we learned how enriching raw log data with contextual information helps validate incidents faster and more accurately.

Importance of a Baseline

Knowing a user's normal login behavior is critical for identifying anomalies. The idea of "context-aware" security was reinforced during our suspicious sign-in analysis.

Framework-Driven Response Enhances Clarity

Applying the NIST framework to our playbook creation gave structure to our thought process. It ensured that no phase of the incident response cycle was overlooked and allowed for clear escalation paths.

Collaboration is Key in SOC Workflows

Almost every task required input from multiple team members triaging logs, analyzing malware, assigning severity levels, and building the playbook. The sprint emphasized communication and shared responsibility.

Security is a Continuous Cycle

From vulnerability management to playbook development, we learned that security isn't a one-off task, it's iterative and must evolve based on lessons from past incidents.

Conclusion

The SOC sprint provided a realistic, hands-on experience of life inside a Security Operations Center. By combining technical tasks like malware analysis and log review with strategic actions like playbook creation and vulnerability risk ranking, I gained a deep appreciation of how cybersecurity professionals prevent, detect, and respond to threats.

Creating the suspicious sign-in playbook using the NIST framework was one of the highlights of the sprint. It required not only technical expertise but also communication, planning, and cross-team input. The experience has solidified my understanding of blue team responsibilities and significantly improved my incident handling skills.